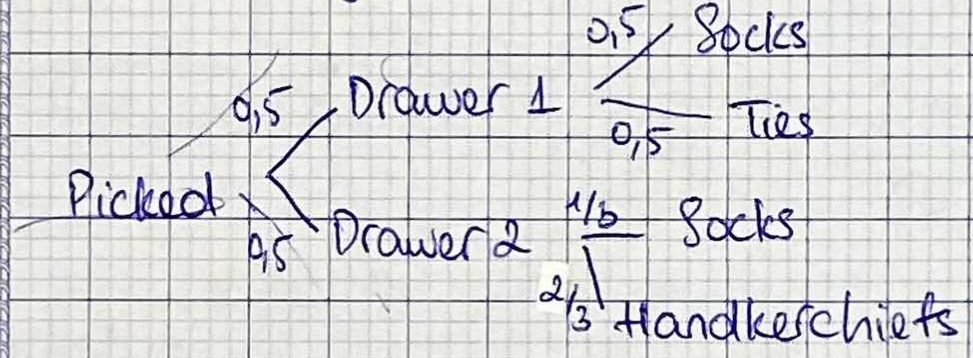
Phú
Hồ Nguyễn Phú

Tutorial 5

- Leader: Hồ Nguyễn Phú
- Cao Tuệ Anh
- Phạm Trọng Quý
- Châu Minh Quân
- Hồng Nguyễn Phúc

---

1. According to the statistics:



We can calculate the probability of picking sock:

$$P(\text{Sock}) = P(\text{Draw } 1) \cdot P(\text{Sock} \mid \text{Draw } 1) + P(\text{Draw } 2) \cdot P(\text{Sock} \mid \text{Draw } 2)$$

$$= 0{,}5 \cdot 0{,}5 + 0{,}5 \cdot \frac{1}{3}$$

$$= \frac{5}{12}$$

TTBOOK

Phu
Hồ Nguyễn Phú

The probability that the item comes from the first draw if turns out to be a sock:

$$P(\text{Draw } 2 \mid \text{Sock}) = \frac{P(\text{Draw } 1 \cap \text{Sock})}{P(\text{Sock})}$$

$$= \frac{P(\text{Draw } 1) \cdot P(S \mid \text{Draw } 1)}{P(\text{Sock})}$$

$$= \frac{0{,}5 \cdot 0{,}5}{5/12} = \frac{3}{5}$$

2)

According to the statistics:

Let H be the event that the person is healthy.

Let Pos be the event that the blood test returns positive result.

Phu
Lò Nguyễn
Phú

$$P(Pos) = P(H) \cdot P(Pos|H) + P(H') \cdot P(Pos|H')$$
$$= 0{,}99 \cdot 0{,}06 + 0{,}01 \cdot 0{,}97$$
$$= 0{,}0691$$

b) $$P(H'|Pos) = \frac{P(H' \cap Pos)}{P(Pos)} = \frac{P(H') \cdot P(Pos|H')}{P(Pos)}$$

$$= \frac{0{,}01 \cdot 0{,}97}{0{,}0691} \simeq 0{,}1404$$

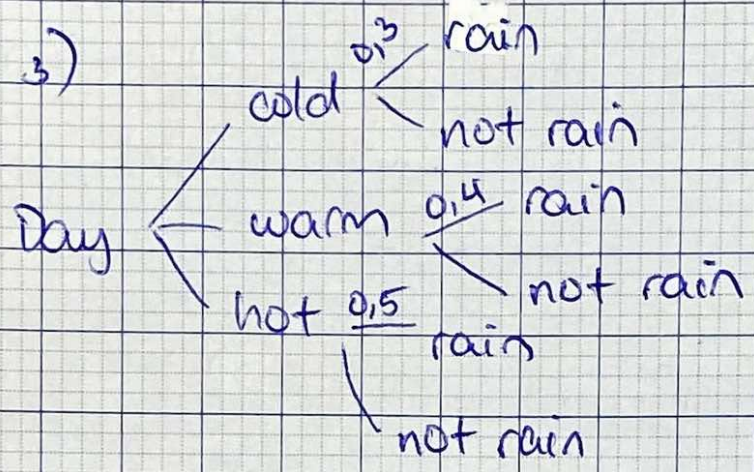
c) $$P(H|Pos') = \frac{P(H \cap Pos')}{P(Pos')} = \frac{P(H) \cdot P(Pos'|H)}{1 - P(Pos)}$$

$$= \frac{P(H) \cdot (1 - P(Pos|H))}{1 - P(Pos)}$$

$$= \frac{0{,}99 \cdot (1 - 0{,}06)}{1 - 0{,}0691} \simeq 0{,}9997$$

Phu
Hồ Nguyễn Phú

3)



$$P(\text{Not rain}) = P(\text{cold}) \cdot P(\text{not rain} \mid \text{cold}) + P(\text{warm}) \cdot P(\text{not rain} \mid \text{warm}) + P(\text{hot}) \cdot P(\text{not rain} \mid \text{hot})$$

$$= 0{,}15 \cdot (1-0{,}3) + 0{,}25 \cdot (1-0{,}4) + (1-0{,}15-0{,}25) \cdot (1-0{,}5)$$

$$= 0{,}555$$

If it is not raining on a particular day, the probability that it is cold.

$$P(\text{cold} \mid \text{not rain}) = \frac{P(\text{cold} \cap \text{not rain})}{P(\text{not rain})}$$

$$= \frac{P(\text{cold}) \cdot P(\text{not rain} \mid \text{cold})}{P(\text{not rain})}$$

$$= \frac{0,15 \cdot (1-0,3)}{0,555} \approx 0,1892$$

4) Let A be the event that the coin is the fair one.
Let A' be the event that the coin is the unfair one.
Let B be the event that the coin falls heads up.

Phu
Hồ Nguyễn
Phú

According to statistics:

We have: $P(A) = P(A') = 0,5$

$P(B|A) = 0,5$

$P(B|A') = \frac{1}{3}$

The probability of the coin falling heads up

$$P(B) = P(A) \cdot P(B|A) + P(A') \cdot P(B|A')$$
$$= 0,5 \cdot 0,5 + 0,5 \cdot \frac{1}{3}$$
$$= \frac{5}{12}$$

The probability that the coin is the fair one if the coin falls heads up:

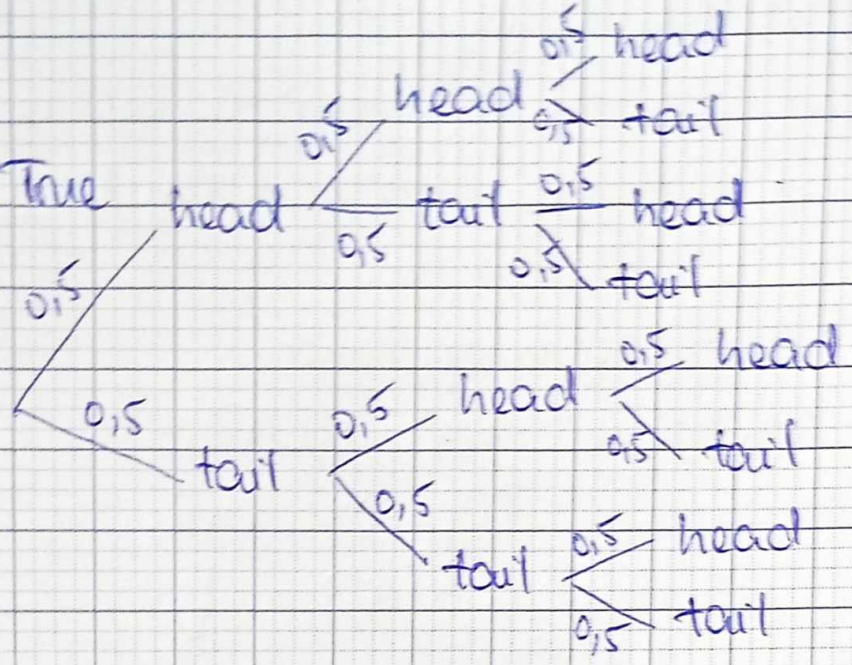
$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B|A)}{P(B)}$$
$$= \frac{0,5 \cdot 0,5}{5/12} = \frac{3}{5}$$

Một trang vở một tương lai
TTBOOK

Phú
Kà Nguyễn
Phú

5)

a. True



P(2 heads, 1 tail) = P(1 head, 2 tails)

$= 0{,}5^3 + 0{,}5^3 + 0{,}5^3 = \frac{3}{8}$

⇒ They are equal to each other.

b) $P(\heartsuit) = \frac{13}{52} = \frac{1}{4}$

$P(A) = \frac{4}{52}$

$P(\heartsuit | A) = \frac{P(\heartsuit \cap A)}{P(A)} = \frac{1/52}{4/52} = \frac{1}{4}$

Phu

Hồ Nguyễn Phú

because $P(\heartsuit \mid A) = P(\heartsuit)$, the probability that the card is a heart does not increase when it is conditioned on the knowledge of that is an ace

$\Rightarrow$ False

Phu
Hồ Nguyễn
Phú

c) True. Because if two events are independent then: $P(A|B) = P(A)$

We have: $P(A \cap B) = P(A|B) \cdot P(B)$

$= P(A) \cdot P(B)$

d) True. Because $P(A) + P(A') = 1$

Similar to $P(A|B) + P(A'|B) = 1$

e) False. Because if they are independent then:

$$\frac{P(A \cap B)}{P(B)} + \frac{P(A \cap B')}{P(B')} = \frac{P(A) \cdot P(B)}{P(B)} + \frac{P(A) \cdot P(B')}{P(B')}$$

$$= P(A) + P(A) \neq 1 \ (\exists A)$$

f) False. Because if two event A and B support each other, they can be: $P(A|B) > P(A)$